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SK hynix Completes World's First HBM4 Development and Readies Mass Production

September 12, 2025



News Highlights

- The company is ready to supply the best-in-class HBM4 to customers according to their plans and to maintain competitive position
- The product's bandwidth has doubled and power efficiency has improved 40% compared to the previous generation
- Company to overcome the technological challenges of the AI era with HBM4, a symbolic turning point beyond AI infrastructure limitations

Seoul, September 12, 2025

SK hynix Inc. (or "the company", www.skhynix.com) announced today that it has completed development and finished preparation of HBM4¹, a next generation memory product for ultra-high performance AI, mass production for the world's first time.

¹**HBM(High Bandwidth Memory)**: This high-value, high-performance memory vertically interconnects multiple DRAM chips and dramatically increases data processing speed in comparison to conventional DRAM products. There are six generations of HBM, starting with the original HBM which was followed by HBM2, HBM2E, HBM3, HBM3E, and HBM4.





SK hynix said that the company has successfully completed development and based on this technological achievement, the company has prepared HBM4 mass production to lead the AI era. Through this momentum, the company has once again proven the AI memory leadership in the global market.

“Completion of HBM4 development will be a new milestone for the industry,” Joohwan Cho, Head of HBM Development at SK hynix, who has led the development, said. “By supplying the product that meets customer needs in performance, power efficiency and reliability in timely manner, the company will fulfill time to market and maintain competitive position.”

With recent dramatic increase in AI demand and data processing, the needs for high bandwidth² memory for faster system speed are surging. In addition, securing memory power efficiency has emerged as a key requirement for customers as power consumption for data center operation has increased. SK hynix expects HBM4, with increased bandwidth and power efficiency, to be the optimal solution to meet customers’ needs.

²**Bandwidth:** In HBM products, bandwidth refers to the total data capacity that one HBM package can process per second.

HBM4, of which mass production system has been readied, has the industry’s best data processing speed and power efficiency with the bandwidth doubled through adoption of 2,048 I/O terminals, double from the previous generation, and power efficiency improved by more than 40%. The company expects to improve AI service performance by up to 69% when the product is applied, which will lead to solve data bottleneck and significantly reduce data center power costs.

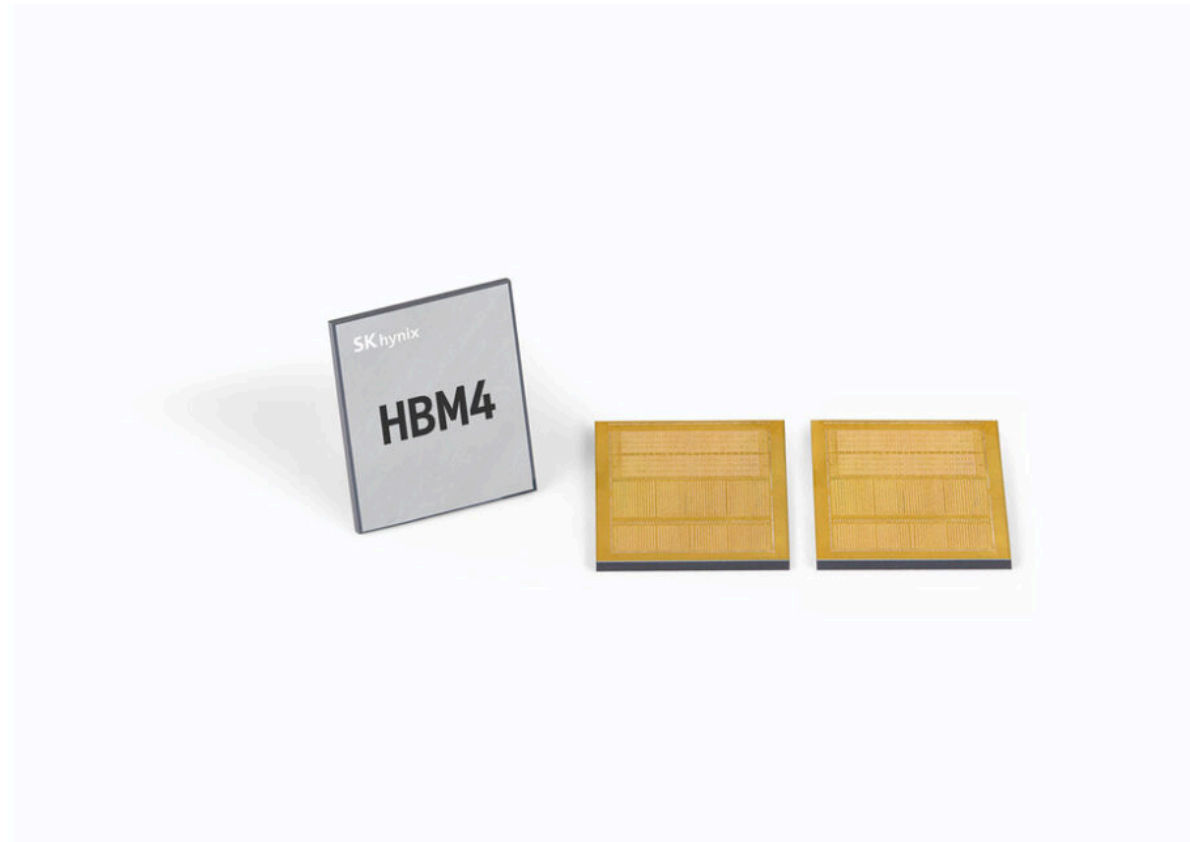
The company has far exceeded the JEDEC³ standard operating speed(8Gbps) by implementing over 10Gbps(Gigabit per second) operating speed in HBM4.

³**JEDEC(Joint Electron Device Engineering Council):** A standardization body that is the global leader in developing open standards and publications for the microelectronics industry.

In addition, the company adopted the Advanced MR-MUF⁴ process in HBM4, which has been proven to be reliable in the market, and the 1bnm process, or the fifth-generation of the 10-nanometer technology, to minimize the risk in mass production.

⁴**MR-MUF(Mass Reflow Molded Underfill):** The process of stacking semiconductor chips, injecting liquid protective materials between them to protect the circuit between chips, and hardening them. The process has proved to be more efficient and effective for heat dissipation, compared with the method of laying film-type materials for each chip stack. SK hynix's advanced MR-MUF technology is critical to securing a stable HBM mass production as it provides good warpage control and reduces the pressure on the chips being stacked.

"We are unveiling the establishment of the world's first mass production system of HBM4," Justin Kim, President & Head of AI Infra at SK hynix, said. "HBM4, a symbolic turning point beyond the AI infrastructure limitations, will be a core product for overcoming technological challenges. We will grow into a full-stack AI memory provider by supplying memory products with the best quality and diverse performance required for the AI era in a timely manner."



About SK hynix Inc.

SK hynix Inc., headquartered in Korea, is the world's top tier semiconductor supplier offering Dynamic Random Access Memory chips ("DRAM") and flash memory chips ("NAND flash") for a wide range of distinguished customers globally. The Company's shares are traded on the Korea Exchange, and the Global Depository shares are listed on the Luxembourg Stock Exchange. Further information about SK hynix is available at www.skhynix.com, news.skhynix.com.

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